

RISC-V core selection for FPGA-based ECG SoC

"Research the best open-source RISC-V core for an Artix-7 FPGA ECG SoC. Compare Ibex, CV32E40P, VexRiscv, NEORV32, CVA6, PicoRV32 and others by ISA, LUT/FF/BRAM/DSP, Fmax, pipeline, interrupts and Artix-7 results. Study ECG/EEG/PPG FPGA systems, AFE SPI timing, FIFO/DMA, UART, timers and interrupts. Recommend the optimal core and full AFE→SPI→RISC-V→buffer→DSP/accelerator→UART architecture using verified GitHub/paper sources." *Generated by LeapSpace*

Artix-7 ECG SoC RISC-V Core Selection at a Glance

Key Findings Table

Topic	Key finding	Most relevant population/system	Evidence
Best-fit core class	Small MCU-style RV32 cores fit ECG SoCs better than large application-class cores because biosignal acquisition is control-dominant and DSP is typically offloaded to hardware.	Artix-7 biomedical acquisition SoCs	[1] [2] [3] [4] [5] [6]
Best named core recommendation	Ibex is the most defensible choice from the supplied evidence because it is explicitly documented as a compact 2-stage FPGA-friendly core, whereas CVA6 is heavier and Artix-7 evidence for CV32E40P, VexRiscv, and NEORV32 is incomplete.	Artix-7 ECG control-plane CPU	[2] [6] [7] [8] [9] [10] [11] [12]
Benchmark limitation	There is no single verified apples-to-apples Artix-7 table across Ibex, CV32E40P, VexRiscv, NEORV32, CVA6, and PicoRV32 with uniform LUT/FF/BRAM/DSP/Fmax/interrupt data.	Cross-core comparison methodology	[2] [6] [11] [12] [13]

Topic	Key finding	Most relevant population/system	Evidence
Interrupt importance	Streaming biosignal systems depend heavily on predictable timer/data-ready interrupt behavior; CLIC/fast IRQ ideas are attractive, but reproducible Artix-7 evidence is sparse.	ECG/EEG/PPG acquisition loops	[14] [15] [16] [17] [18]
Architecture pattern	The strongest recurring system pattern is AFE/ADC DRDY -> SPI capture -> FIFO/DMA -> CPU control plane -> hardware DSP/ML -> host I/O .	Biomedical FPGA systems	[5] [6] [19] [20] [21] [22]
Sampling pressure on CPU	ECG-class systems in the supplied evidence mostly operate in the hundreds of Hz to low kHz range, reducing pressure on CPU throughput and increasing the value of buffering and interrupt reliability.	ECG/EEG/PPG acquisition	[19] [23] [24] [25] [26]
UART role	UART is suitable for debug, control, and low-rate telemetry; continuous raw-stream robustness depends on FIFO buffering and preferably DMA-backed buffering rather than CPU polling.	FPGA SoC host link	[6] [20] [21] [27]
DSP partitioning	Prior systems consistently keep hot-path filtering, FFT, QRS/CNN, or feature extraction in hardware while firmware handles setup, framing, and communications.	Biomedical CPU/accelerator partitioning	[5] [22] [28] [29]

Direct Answer

For an Artix-7 FPGA ECG SoC, the strongest evidence-based choice is **lbex as the control-plane CPU**, paired with **hardware SPI capture, FIFO/ping-pong BRAM buffering, optional DMA, and hardware DSP/accelerators for filtering or feature extraction** rather than relying on the CPU for signal processing [2] [5] [6] [7] [20].

Why Ibex, not the others? The supplied evidence explicitly places Ibex in the small 2-stage, FPGA-friendly class, including one Artix-7-related report with 3841 LUTs and 2030 FFs and another Zynq-7000 implementation at 125 MHz with 4591 LUTs, 3619 FFs, and 256 BRAM cells [2] [7]. CVA6 is represented as a higher-complexity core without supporting small-Artix-7 evidence, making it poorly matched to a compact ECG SoC [8] [9] [10]. CV32E40P has appealing interrupt features, including reported CLIC/fastirq support down to 6 cycles, but the supplied evidence does not verify comparable Artix-7 implementation data for this use case [11] [14]. VexRiscv and NEORV32 likewise lack a verified Artix-7 comparison in the provided material [6] [12].

Recommended architecture:

AFE/ADC with DRDY -> FPGA SPI master -> small async FIFO -> optional DMA or ping-pong BRAM -> Ibex firmware for control, framing, error handling, timestamps, and interrupts -> hardware DSP/accelerator for FIR/IIR/FFT/QRS/features -> UART for debug/commands and low-rate telemetry [6] [13] [19] [20] [21] [22].

Taken together, these findings suggest that for an ECG SoC, **buffering and autonomous movement matter more than raw CPU performance**, so the optimal design is a small, easy-to-integrate RISC-V core plus deterministic peripherals and hardware offload [6] [19] [20] [21].

Confidence level: Medium-High - the architectural recommendation is strongly supported, but the named-core ranking is limited by fragmented Artix-7 benchmark data [2] [6] [11] [12].

Study Scope

- **Period covered by supplied evidence:** mainly 2018-2025 style FPGA/RISC-V and biomedical system reports as represented in the retrieved excerpts [7] [13] [14] [20] [26].
- **Disciplines covered:** RISC-V microarchitecture, FPGA SoC design, biomedical acquisition, SPI/ADC interfacing, buffering/DMA, and hardware acceleration [6] [13] [19] [30].
- **Methods covered:** FPGA synthesis/timing excerpts, architecture papers, SoC implementation reports, and biomedical case studies [23] [31] [32] [33] [34].

Assumptions & Limitations

- The supplied evidence does **not** provide a unified, reproducible Artix-7 benchmark table for all requested named cores, so recommendation must rely on architecture fit plus partial implementation evidence rather than strict side-by-side benchmarking [2] [6] [11] [12] [13].
- Exact SPI CPOL/CPHA, burst length, FIFO depth, DMA descriptor formats, UART throughput ceilings, and interrupt latency for a complete open-source Artix-7 ECG RISC-V reference design are not verified in the supplied material [18] [35] [36] [37].
- CV32E40P's strong interrupt story is conceptually attractive, but board-level Artix-7 evidence in the supplied sources is not sufficient to outrank Ibex with confidence [11] [14].

Suggested Further Research

- Run a controlled **XC7A35T/XC7A100T** comparison of Ibex, CV32E40P, VexRiscv, and NEORV32 with identical peripherals, memory map, and firmware workload [6] [7] [38].
- Measure real **interrupt latency and jitter** under simultaneous SPI capture and UART transmission on Artix-7 [14] [17] [39].
- Publish a full open ECG reference design with verified **DRDY timing, FIFO depth, DMA policy, and sustained throughput** [6] [19] [37].
- Compare **polling vs interrupt vs DMA** for sub-kHz single-channel and multi-channel biosignal workloads [6] [17] [20].
- Evaluate whether feature-only UART streaming is sufficient while raw data is retained locally in BRAM/DDR for burst export [21] [25] [27].

Full Report Body

1. Foundational Context

An ECG SoC on Artix-7 is not primarily a compute-throughput problem; it is a **deterministic acquisition and data-movement problem**. The CPU must configure the AFE/ADC, respond to timer or DRDY events, supervise SPI capture, manage buffering, handle exceptions/interrupts, and package data or extracted features for output [6] [17] [19]. The hot path—filtering, FFT, QRS detection, CNN inference, or feature extraction—is repeatedly moved into hardware in prior biomedical systems [5] [13] [22] [28].

This distinction matters for core choice. Small RV32I/RV32IM-class processors are repeatedly presented as the efficient FPGA option, while larger pipelines and richer memory systems incur area/frequency penalties not justified by control-heavy workloads [1] [2] [3] [4]. In other words, the CPU should be a **supervisor**, not the main signal processor [5] [6].

Confidence level: High - this pattern is consistent across RISC-V and biomedical case-study evidence [5] [6] [22].

2. RISC-V Core Comparison for Artix-7

2.1 What can be compared reliably

The supplied evidence supports comparison on:

- ISA class and complexity trends [30] [40] [41]
- Pipeline depth and control-vs-performance tradeoffs [1] [42] [43]
- Partial FPGA implementation evidence for some cores or comparable RV32 designs [2] [31] [32] [33] [44]
- Interrupt architecture concepts and availability [14] [15] [16]

It does **not** support a complete verified Artix-7 LUT/FF/BRAM/DSP/Fmax ranking across all named cores [2] [6] [11] [12].

2.2 Candidate-by-candidate assessment

Core	Evidence from supplied material	Implication for ECG SoC
Ibex	Explicitly identified as a 2-stage small core; one report gives 3841 LUTs and 2030 FFs; another Zynq-7000 implementation reports 4591 LUTs, 3619 FFs, 256 BRAM cells, 125 MHz [2] [7]	Strong fit for compact control-plane role
CV32E40P	MCU-class core; CLIC + fastirq can reduce latency to 6 cycles, but no verified Artix-7 implementation metrics in supplied evidence [11] [14]	Architecturally attractive, but insufficient board evidence
VexRiscv	Mentioned only as a comparison point; no supplied Artix-7 plugin/resource table [12] [45]	Cannot rank conclusively here
NEORV32	Absent from retrieved papers [46]	No evidence-based ranking possible
CVA6	Appears only in higher-complexity performance/area discussions; no small-Artix-7 evidence [8] [9] [10]	Likely over-scaled for ECG control plane
PicoRV32	Resource-efficient in a Zynq-7000 design; 5,208 slice LUTs and 100 MHz timing closure reported, but not isolated as a full core-only breakdown [44]	Viable minimalist option, but less complete interrupt evidence here

2.3 ISA and pipeline tradeoffs

Most retrieved FPGA-oriented designs sit in the RV32I/RV32IM space, with single-cycle through six-stage organizations [3] [4] [30] [40] [41] [47] [48] [49]. The M extension raises resource cost and can reduce Fmax, although some RV32IM designs still achieve practical FPGA results [41] [50]. Five-stage pipelines improve throughput relative to single-cycle designs, but add hazard, forwarding, and stall complexity [3] [4] [30]. Smaller 2-stage cores are explicitly identified as the smallest options in the supplied set [1] [2].

For ECG control, these tradeoffs favor:

- **RV32I or light RV32IM** over complex application-class profiles [1] [4]
- **2- to 3-stage simplicity** over deeper pipelines unless firmware compute is unexpectedly heavy [2] [42]
- **predictable interrupt/trap handling** over peak benchmark speed [15] [17]

2.4 Interrupt and exception capability

Interrupt behavior matters disproportionately because biosignal capture is often periodic or DRDY-triggered [17] [19]. The supplied sources distinguish local interrupts handled by CLINT and prioritized external interrupts via PLIC [15]. CLIC is described as a lower-latency alternative with improved vectored jumps and tail-chaining, but many open-source cores still do not support it in practice [16]. CV32E40P is the clearest named core in the supplied evidence with explicit fast interrupt claims, reportedly reaching 6-cycle latency with CLIC/fastirq support [14].

Taken together, these findings suggest:

- If verified low-latency interrupt implementation is the overriding requirement, **CV32E40P is conceptually compelling** [14].
- If verified Artix-7 fit and overall conservatism dominate, **Ibex remains the safer recommendation** because its small-core FPGA role is better evidenced in the supplied material [2] [7].

2.5 Practical selection conclusion

Recommended core: Ibex.

Reasoning:

1. It is explicitly evidenced as a compact 2-stage core on FPGA [2].
2. ECG workloads mainly need deterministic control, not Linux-class compute [5] [6].
3. CVA6 is too heavy for this role [8] [9] [10].
4. The supplied Artix-7 evidence for CV32E40P, VexRiscv, and NEORV32 is too incomplete for a stronger verified recommendation [6] [11] [12].

Confidence level: Medium - high confidence in the small-core-vs-large-core conclusion; medium confidence in choosing Ibex over CV32E40P/VexRiscv due to incomplete like-for-like data [2] [11] [14].

3. Biomedical FPGA SoC Case Studies

The supplied biomedical evidence strongly supports **hardware/CPU partitioning**.

- QRS/beat detection is often done in hardware, with higher-level logic on an embedded processor [22].
- In a Zynq-7000 ECG CNN system, hardware handles preprocessing/classification while a RISC-V microprocessor handles transmission and control [5] [28].
- An Artix-7 RISC-V SoC with SPI and FFT acceleration ran at 48 MHz and 108 mW, showing the feasibility of pairing a soft CPU with DSP hardware [13].
- An Artix-7 open-source RISC-V SoC with UART achieved 1.09 DMIPS/MHz at 50 MHz, demonstrating modest but sufficient embedded-control performance [49].

For integration analogs:

- A PYNQ-Z1 EEG platform used ADS1299 via SPI daisy-chain and performed bandpass filtering and FFT in FPGA logic [51].
- A Spartan-6 cardiac monitor used three MicroBlaze cores split across HMI, DSP, and master control, reinforcing functional partitioning by task type [34].

This cross-source pattern implies that an ECG SoC should not attempt to make the RISC-V core the primary DSP engine [5] [6] [13].

Confidence level: High - consistent across multiple biomedical case studies [5] [22] [51].

4. AFE, SPI, and Acquisition Timing

4.1 AFE requirements

ECG AFEs in the supplied evidence emphasize low-noise gain, DC suppression, and bandwidth shaping to preserve weak biosignals under practical electrode offsets and low-power operation [52] [53] [54] [55] [56].

Representative signal-chain settings include:

- 512 Hz sampling with 0.5 Hz high-pass and 200 Hz low-pass in a wearable ECG patch [24]
- 853 Hz acquisition in an ADS1293-based board with 0.5 Hz high-pass and 50 Hz low-pass filtering [23]
- AD8232-based chain with 0.5 Hz high-pass, 40 Hz low-pass, and anti-aliasing cutoff at Nyquist [57] [58]
- Simultaneous ECG/PPG at 1000 Hz [25]

These rates are modest relative to FPGA clock rates, reinforcing that reliable scheduling and buffering matter more than CPU horsepower [19] [23] [25].

4.2 SPI timing and handshakes

The strongest acquisition-interface evidence is architectural rather than protocol-specific:

- FPGA logic can generate the sampling clock and use ADC conversion-complete/DRDY to begin SPI readout into a buffer [19].
- FPGA SPI interface modules may use DCM-generated SCLK at multiple speeds [35].
- Data-ready/frame-sync style handoff to downstream host or storage is common [36].

The supplied snippets do **not** verify exact CPOL/CPHA or burst length settings [19] [35] [36] [59].

Therefore, the defensible design recommendation is to center the architecture on **DRDY-triggered SPI transactions with buffering immediately after capture** [19] [21].

4.3 Practical implication for ECG SoC timing

A robust Artix-7 ECG path should use:

1. AFE/ADC in continuous mode or timed conversion mode [23] [24]
2. DRDY or conversion-complete into FPGA control logic [19]
3. SPI burst into a small FIFO crossing into the system clock domain [35] [60]
4. Half/full-buffer interrupt or DMA completion to firmware [29] [61]

Confidence level: High - the event-driven SPI/buffered pattern is directly supported, even though low-level SPI mode details are not [19] [21].

5. Buffering, DMA, UART, Timers, and Offload

5.1 FIFO and buffering

FIFO buffering is repeatedly used to absorb burstiness and decouple producer/consumer timing in UART and data-stream systems [21] [27]. FPGA systems also commonly rely on embedded memory blocks and FIFO structures for buffering and DSP slices for acceleration [60].

For an ECG SoC, this implies:

- a **small capture FIFO** immediately after SPI,
- a **larger ping-pong BRAM buffer** for firmware-visible blocks,
- optional **DMA** when the system includes more than trivial streaming or acceleration [6] [20].

5.2 DMA role

The most relevant SoC evidence shows a RISC-V CPU orchestrating accelerator and DMA configuration while data moves autonomously after launch [6] [29] [61] [62]. Scatter-gather or programmable DMA approaches reduce CPU intervention and keep data movers busy [20] [63].

For ECG-class streaming, a sensible policy is:

- CPU sets descriptors or transfer length,
- DMA moves FIFO/BRAM blocks to accelerator input or output region,
- interrupt fires only on half/full block completion or error [29] [62].

5.3 UART use

The supplied UART evidence emphasizes buffering and robustness rather than high throughput [21] [27]. Since continuous raw biosignal streams can create backpressure, UART should be treated as:

- debug/programming/control path,

- low-rate telemetry path,
- raw-stream path only for very low channel counts or lower duty-cycle exports [21] [25] [27].

5.4 Timers and interrupts

Interrupt-driven acquisition is well aligned with sensor sampling [17]. FPGA timing controllers can be built from counters generating periodic triggers and address patterns [64]. In practice, ECG systems should combine:

- a timer for periodic sampling if ADC is host-driven,
- DRDY interrupt if ADC is self-timed,
- watchdog/error flag for FIFO overflow, missed frame, or UART congestion [17] [21] [64].

5.5 DSP/accelerator placement

The most supported placement is:

- CPU: register programming, control, framing, packetization, state management [5] [6]
- Hardware: FIR/IIR, FFT, QRS, feature extraction, possibly lightweight CNN/inference [5] [13] [22] [65]

Confidence level: High - this partitioning is one of the clearest patterns in the supplied evidence [5] [6].

6. Recommended Full Architecture

6.1 Proposed data path

AFE/ADC -> DRDY -> SPI master -> async/sample FIFO -> ping-pong BRAM -> Ibex control core -> DMA/stream switch -> DSP/accelerator -> output buffer -> UART

6.2 Block-by-block rationale

- **AFE/ADC:** provides conditioned ECG samples with application-appropriate high-pass/low-pass behavior and low-noise front-end design [23] [52] [57].
- **DRDY + SPI master:** deterministic capture tied to conversion-complete signaling [19].
- **Async FIFO:** absorbs clock-domain differences and prevents CPU-service jitter from causing sample loss [21] [60].
- **Ping-pong BRAM:** allows one block to fill while another is processed or transmitted [6] [60].
- **Ibex CPU:** configures peripherals, handles interrupts, timestamps blocks, manages faults, and supervises DMA/accelerators [2] [6].
- **DMA/stream switch:** reduces firmware load and decouples acquisition from compute/output [20] [29].
- **DSP/accelerator:** performs filtering/FFT/QRS/features without burdening the CPU [13] [22].

- **UART:** command/debug/output path for features or low-rate telemetry [21] [27].

6.3 Minimal and expanded variants

Minimal single-channel ECG SoC

- SPI -> FIFO -> Ibex -> BRAM -> UART
- Use interrupts, no DMA, basic digital filtering in hardware only if needed [19] [21]

Recommended scalable SoC

- SPI -> FIFO -> ping-pong BRAM -> DMA -> FIR/FFT/QRS accelerator -> output FIFO -> UART
- Ibex wakes on half/full buffer and error conditions [6] [13] [20]

6.4 Final recommendation

If forced to pick one open-source RISC-V core today from the supplied evidence, choose **Ibex** for the Artix-7 ECG SoC, not because it is proven fastest, but because it is the most clearly evidenced **small, FPGA-appropriate, control-plane-compatible** option in the available material [2] [7]. Pair it with **buffer-first acquisition, event-driven SPI, optional DMA, and hardware DSP offload** [6] [13] [19] [20].

Overall conclusion confidence: Medium-High - the system architecture recommendation is strong and well supported; the precise named-core superiority claim remains limited by fragmented Artix-7 benchmark coverage [2] [6] [11] [12].

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